

BUREAU OF INDIAN STANDARDS**DRAFT FOR COMMENTS ONLY***(Not to be reproduced without permission of BIS or used as an Indian Standard)**भारतीय मानक मसौदा***चिकित्सा पद्धति में विद्युत उपकरणों के लिए ग्राफ़िकल प्रतीक***[IS 14180 : 2019/ IEC 60878 : 2015 का दूसरा पुनरीक्षण, संशोधित]**Draft Indian Standard***Graphical symbols for electrical equipment in medical practice***[Second Revision of IS 14180: 2019/ IEC 60878: 2015, MOD]***ICS 01.080.40, 11.040.60**

Electromedical, Diagnostic Imaging and Radiotherapy Last date for comments: **28 August 2026**
Equipment Sectional Committee, MHD 15

NATIONAL FOREWORD*(Adoption clause will be added later)*

Corrigendum 1 published in 2023 to IEC TR 60878:2022 is given at the end of this publication.

This standard was first published as IS 14180: 2019/ IEC 60878: 2015 which was an identical adoption of IEC TR 60878:2015 ‘Graphical symbols for electrical equipment in medical practice’. The second revision has been brought out to align the Indian Standard with the consolidated version of IEC TR 60878:2022 ‘Graphical symbols for electrical equipment in medical practice’.

The text of IEC standard has been approved as suitable for publication as an Indian Standard without deviations. Certain conventions are however not identical to those used in Indian Standards. Attention is particularly drawn to the following:

- a. Wherever the words ‘International Standard’ appear referring to this standard, they should be read as ‘Indian Standard’; and
- b. Comma (,) has been used as a decimal marker while in Indian Standards, the current practice is to use a point (.) as the decimal marker.

In this adopted standard, reference appears to certain International Standards for which Indian Standard also exists. The corresponding Indian Standards, which are to be substituted in their respective places, are listed below along with their degree of equivalence for the editions indicated:

*International Standard**Corresponding Indian
Standard**Degree of
Equivalence*

IEC 60417, Graphical symbols for use on equipment	IS 16450: 2023/ISO 7000:2019, Graphical Symbols for Use on Equipment — Registered Symbols (First Revision)	Identical
ISO 7000, Graphical symbols for use on equipment – Registered symbols		
IEC 60601-1, Medical electrical equipment – Part 1: General requirements for basic safety and essential performance	IS 13450 (Part 1): 2024, Medical Electrical Equipment Part 1 General Requirements for Basic Safety and Essential Performance (IEC 60601-1: 2020, MOD) (Third Revision)	Modified
IEC 60601-1-8:2006, Medical electrical equipment – Part 1-8: General requirements for basic safety and essential performance – Collateral Standard: General requirements, tests and guidance for alarm systems in medical electrical equipment and medical electrical systems	IS 13450 (Part 1/Sec 8): 2024, Medical Electrical Equipment Part 1 General Requirements for Basic Safety and Essential Performance Section 8 General Requirements, Tests and Guidance for Alarm Systems in Medical Electrical Equipment and Medical Electrical Systems (IEC 60601-1-8 : 2020, MOD) (First Revision)	Modified
IEC 60601-1-8:2006/AMD1:2012 IEC 60601-1-8:2006/AMD2:2020		
IEC 60601-2-18:2009, Medical electrical equipment – Part 2-18: Particular requirements for the basic safety and essential performance of endoscopic equipment	IS 13450 (Part 2/Sec 18) : 2014 IEC 60601-2-18 : 2009, Medical Electrical Equipment Part 2 Particular Requirements for the Basic Safety and Essential Performance Section 18 Endoscopic Equipment	Identical
IEC 60601-2-22:2019, Medical electrical equipment – Part 2-22: Particular requirements for basic safety and essential performance of surgical, cosmetic, therapeutic and diagnostic laser equipment	IS 13450 (Part 2/Sec 22) : 2021 IEC 60601-2-22 : 2019, Medical Electrical Equipment Part 2 Particular Requirements for Basic Safety and Essential Performance Section 22 Surgical, cosmetic, therapeutic and diagnostic laser equipment (First Revision)	Identical

IEC 61140, Protection against electric shock – Common aspects for installation and equipment	IS 9409 : 2023/ IEC 61140 : 2016, Protection Against Electric Shock — Common Aspects for Installation and Equipment (<i>First Revision</i>)	Identical
IEC 62056-21, Electricity metering – Data exchange for meter reading, tariff and load control – Part 21: Direct local data exchange	IS/IEC 62056-21 : 2002, Electricity Metering — Data Exchange for Meter Reading, Tariff and Load Control Part 21 Direct Local Data Exchange	Identical
ISO 361, Basic ionizing radiation symbol	IS 16885 : 2018/ ISO 361 : 1975, Basic Ionizing Radiation Symbol	Identical
ISO 3166-1, Codes for the representation of names of countries and their subdivisions – Part 1: Country codes	IS 14836 (Part 1):2020/ ISO 31066-1 : 2013, Codes for the Representation of Names of Countries and their Subdivisions Part 1 Country Codes (<i>Second Revision</i>)	Identical
ISO 3864-1, Graphical symbols – Safety colours and safety signs – Part 1: Design principles for safety signs and safety markings	IS 16449 (Part 1) : 2018/ ISO 3864-1 : 2011, Graphical Symbols — Safety Colours and Safety Signs Part 1 Design Principles for Safety Signs and Safety Markings	Identical
ISO 7001, Graphical symbols – Public information symbols	IS 15504 : 2023/ ISO 7001 : 2023, Graphic Symbols — Registered Public Information Symbols (<i>Second Revision</i>)	Identical
ISO 7010, Graphical symbols – Safety colours and safety signs – Registered safety signs	IS 16451 : 2023/ ISO 7010 : 2019, Graphical Symbols — Safety Colours and Safety Signs — Registered Safety Signs (<i>First Revision</i>)	Identical
ISO 8601-1, Date and time – Representations for information interchange – Part 1: Basic rules	IS/ISO 8601-1 : 2019, Date and Time — Representations for Information Interchange Part 1 Basic Rules	Identical
ISO 15223-1:2021, Medical devices – Symbols to be used with information to be supplied by the manufacturer – Part 1: General requirements	IS 18105 (Part 1) : 2023/ ISO 15223-1 : 2021, Medical Devices — Symbols to be Used with Information to be Supplied by the Manufacturer	Identical

Part 1 General Requirements
(third revision)

The technical committee has reviewed the provisions of the following International Standards referred in this adopted standard and has decided that these are acceptable for use in conjunction with this standard:

<i>International Standard</i>	<i>Title</i>
IEC 60601-2-83:2019	Medical electrical equipment – Part 2-83: Particular requirements for the basic safety and essential performance of home light therapy equipment
IEC 60825-1:2014	Safety of laser products – Part 1: Equipment classification and requirements
IEC 62570:2014	Standard practice for marking medical devices and other items for safety in the magnetic resonance environment

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed or calculated, expressing the result of a test or analysis shall be rounded off in accordance with IS 2: 2022 ‘Rules for rounding off numerical values (*second revision*)’. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Scope

This document provides a compilation, for easy reference, of graphical symbols (graphics, title, description) and safety signs for medical electrical equipment. The graphical symbols are grouped in sections according to their specific field of application (see Clause 4)

The technical content of the document has not been enclosed as it is identical with the corresponding IEC standard. For details, please refer to **IEC TR 60878:2022** or kindly contact:

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